

Exploring AI Frame Interpolation Techniques for Watercolor Animation

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Introduction

The animated short film *Sensual* explores a novel workflow for hand-painted watercolor animation, integrating AI-based frame interpolation techniques to enhance efficiency while preserving artistic integrity. Traditional hand-painted animation is labor-intensive, requiring manual creation of inbetween frames. We investigated AI-driven methods to bridge the gap between artistry and automation.

Evaluation

We evaluated various interpolation models for their suitability in maintaining the organic quality of watercolor animation (see comparison).

As a base for the tests, one shot was animated as a 2D outline animation, then every third frame was printed and painted with watercolor. We either used the 2D outline animation to guide the interpolation, or interpolated between the painted images directly. We noticed three interpolation issues we had to address in our workflow: blurriness, morphing artifacts and pulsing, a rhythmic inconsistency where linear interpolation movements disrupt the non-linear flow of the animation (see supplementary video).

Ultimately we based our workflow on RIFE (Real-Time Intermediate Flow Estimation [1]), due to its high visual consistency, combined with compositing. Its integration in the compositing tool Nuke made it easy to implement into our pipeline. But, like other models, it introduced pulsing artifacts, addressed in our proposed workflow.

Conclusion

By strategically combining AI interpolation with traditional compositing, we achieved a substantial reduction in production time (200 working days to 100 of painting time) without compromising the artistic vision. This workflow can serve as a foundation for future hand-painted animation projects. We believe that AI should act as an augmentative tool for artists rather than a replacement.

References

[1] Huang, Z., Zhang, T., Heng, W., Shi, B., & Zhou, S. (2022). Real-Time Intermediate Flow Estimation for Video Frame Interpolation. arXiv:2011.06294. <https://arxiv.org/abs/2011.06294>

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More information and comparison video:
<https://filmakademiernd.github.io/TD25/>



Comparison



Technique

Pros

Cons

Linear Blending (3)

Efficient blending algorithm, simple implementation

Ghosting, noticeable stuttering

Optical Flow (4)

Estimates motion well, fast interpolation

Artifacts on complex textures & motion

EbSynth (5)

Propagates underlying motion

Needs animated inbetweens, morphing

Diffusion Models (6)

High-quality image generation

Inconsistent results, pulsing, copyright

ABME (7)

Good interpolation of live-action footage

No arbitrary fps, loss of detail, pulsing

Google FILM AI (8)

Good interpolation for live action and stylized footage

Texture warping artifacts in motion, pulsing

Topaz Video AI (10)

Fewer morphing artifacts compared to other methods

Increased blurriness, loss of fine detail

Topaz + EbSynth (11)

Restores some texture details

Low resolution and color depth

RIFE (9)

Good interpolation & temporal accuracy

Morphing artifacts, pulsing

RIFE + Compositing (12)

Enhanced control over interpolation via simulated flickering; preserves style

Requires additional manual compositing; increased workflow complexity

Proposed Workflow

We realized that reducing framerate (8fps instead of 12fps), and reducing from two to one inbetweens, helped to mitigate pulsing and led to a more fluid perception of movement.

1. Animation and Painting: Every third frame is hand-painted after outline animation.
2. Denoise and stabilization
3. Interpolation: RIFE generates inbetween frames, randomly switching between three variants: default (A), mirrored (B) and altered settings (C)
4. Compositing Enhancements: Random transforms, flickering, brush and paper texture

